MILITARY SPECIFICATION

DISHWASHING MACHINE, COMMERCIAL (SUBMARINE USE)

1. SCOPE

1.1 This specification covers an undercounter or flush counter top type dishwashing machine for uniformly washing and rinsing dishes in racks by spraying an aqueous detergent solution and rinse water from above and below and for rendering eating utensils physically and hygienically clean and in such sanitary conditions as to prevent transmission of disease. The machine shall be capable of being readily disassembled and reassembled for submarine installation (see 3.4).

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein.

STANDARD

MIL-STD-761 - Electric Power, Alternating Current, For Shipboard Use, Characteristics and Utilization of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, N.Y. 10017)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016)

NATIONAL SANITATION FOUNDATION

Standard Number 3 - Spray-Type Dishwashing Machines.

(Application for copies should be addressed to the National Sanitation Foundation, Ann Arbor, Michigan).

FSC 7320

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
Standard 45 - Recommended Practice for Electric Installations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Box A, Lenox Hill Station, New York 21, New York).

AMERICAN STANDARDS ASSOCIATION, INC.
ASA - A13a-1928 - Scheme for the Identification of Piping Systems.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 Preproduction. - Prior to beginning production one machine shall be submitted for the preproduction tests specified in 4.2.3 to determine whether the dishwashing machine is suitable for submarine application.

3.2 Material. - Tank and body of machine shall be stainless steel, AISI types 302 or 304, having a commercial No. 4 finish, or better, on exterior surfaces. All other parts (manifolds, piping, valves, scrap screens, pump impellers, and so forth), shall be made of corrosion resisting metals. Materials used shall be of such a nature, weight and gage, and so applied, as to be resistant to change by denting, buckling, chipping and pitting in the normal operating environment. When welded seams are used, the weld area and deposited weld metal shall be equally as corrosion - resistant as the parent material.

3.3 Design. - Machine shall be of the undercounter or flush counter top type, front loading and unloading, automatically timed to wash and rinse dishes in racks by the spray method from above and below, and shall be electrically heated and operated.

3.3.1 Size. - Machine shall be suitable for mounting under or flush with a dresser top having the top surface located not more than 37 inches above the finished deck. Body of machine, exclusive of fittings and attachments, shall not exceed 24 inches in width and depth or, in the case of a round machine, shall not exceed a diameter of 24 inches.

3.3.2 Capacity. - Machine shall be capable of handling approximately 40 racks of dishes per hour. Racks used in the machine shall each accommodate, without masking, not less than seven dinner plates or fifteen cups.

3.3.3 Timing mechanism. - The timing mechanism shall be electric of reliable quality to provide not less than a 60 second wash phase, approximate 3 second lag, and not less than a 10 second rinse phase. The total cycle (wash, lag and rinse) shall not exceed 80 seconds. The timing mechanism shall be interconnected with a pilot light to indicate when machine is operating.

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3.3.4 Wash tank. - Wash tank shall be of such capacity or design that pump suction or intake will not become uncovered when machine is tilted 15 degrees in any direction while operating. Water in the wash tank shall be heated by an electric immersion type heater that is thermostatically controlled to maintain wash water temperature between 140°F. and 160°F, or by other suitable means to maintain required wash water temperature.

3.3.5 Rinse water. - A reservoir, built into the machine, or a separate tank, for installation outside of the machine, shall be provided for the rinse water. The reservoir or tank shall be electrically heated and thermostatically controlled to maintain the rinse water temperature between 180°F. and 195°F. The reservoir or tank shall be of such capacity and the electric heaters of such size or number that rinse water will be supplied at required temperature and volume to rinse the dishes at a rate of not less than 40 racks per hour. The reservoir or tank shall be fitted with a thermostatic switch, so wired, that it will de-energize the motor and stop the machine if the temperature of the rinse water drops below 180°F.

3.3.6 Spray tubes or arms. - Two systems of spray tubes or arms shall be provided, one system to spray wash water on the dishes and one to spray rinse water on the dishes. The number of spray tubes or arms provided in each system shall be adequate to form a spray pattern that will saturate the entire area of the dishwashing rack used. Pressure, volume and pattern of wash water, with detergent in proper proportion, shall be such that dishes in a rack will be rendered visually clean within 60 seconds. Pressure, volume and pattern of rinse water shall be such as to sanitize all surfaces of the dishes in a rack in not more than 12 seconds.

3.3.7 Controls. - Controls and gauges used in the operation of the machine shall be contained in a box or console for mounting on the bulkhead in back of the machine. The bottom of the console, or box, will normally be installed about 4 feet 6 inches above the finished deck.

3.3.8 Drain valve and overflow. - The machine shall be fitted with a drain valve and a suitable overflow to prevent the machine from flooding, or a combination drain valve and overflow to accomplish the above may be provided.

3.4 Construction. - The machine shall be so constructed that it can be readily disassembled into sections or parts that can be passed through a 25-inch diameter submarine hatch and door opening having maximum dimensions of 20 inches in width by 38 inches in height, with top and bottom of opening formed on a 10 inch radius; and reassembled with a minimum of effort in the submarine galley or scullery to provide durable and efficient operation. The machine, when assembled, shall be of rigid and durable construction. Base or legs of machine shall be fitted with four holes (one for each leg) for bolting the machine to the deck.

3.4.1 Sanitation. - The machine shall be constructed for ease of cleaning and sanitary aspects to comply with the requirements of National Sanitation Foundation Standard Number 3.

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3.5 Electrical requirements. - Unless otherwise specified in the contract or order (see 6.1), all electrical equipment shall conform to the recommended practice for electrical installation aboard ship as specified by the requirements in IEEE Standard 45. Power supplied to the machine will be 440 volt, 3 phase, 60 cycle, alternating current conforming to the requirements for type I of MIL-STD-761. The manufacturer shall provide a transformer, transformers and or electrical relays as necessary for other specified electrical characteristics. All electrical equipment shall be designed for operation on an ungrounded electrical system but shall operate satisfactorily and without hazard to personnel or equipment with an accidental ground on any conductor. Control equipment shall be splashproof.

3.5.1 Motor. - Motor shall operate the machine under normal operating conditions without exceeding permissible temperature rise and shall have the following characteristics:

Ambient temperature	50°C.
Service	C.
Enclosure	Totally enclosed.
Cooling	Natural or fan.
Horsepower	As required.
Speed	Constant.
Duty	Continuous.
Winding	Fractional horsepower (squirrel-cage induction); Integral horsepower (squirrel-cage induction design B).
Voltage	440 volt-3 phase, 60 cycle or 115 volt, 3 phase, 60 cycle, alternating current for integral horsepower (115 volt, single phase, 60 cycle permitted for fractional horsepower), unless otherwise specified (see 6.1).
Bearings	Ball
Insulation	Class B, or better.

3.5.2 Controllers. - Contactors shall switch all three lines of three phase circuits and both sides of single phase power circuits. All heater, integral motor power circuits shall be 3 phase and all heater and motor power circuits shall be controlled by contactors, which in turn may be actuated by relays. All control circuits shall operate at 115 volts or less. All solenoid valves shall be provided with Epoxy encapsulated coils. The control equipment shall have the following characteristics:

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Ambient temperature	50°C.
Enclosure	Splash proof.
Operation	Magnetic.
Type	Across line (A.C.).
Function	Motor and heater starting.
Duty	Continuous.
Performance	Semi-automatic. Automatic thermostatic cut-off on low rinse water temperature.
Protection	Thermal overload protection and low voltage protection (control circuit to be fused).

3.5.3 Thermostatic switches. - Thermostatic switches shall be of durable construction and capable of controlling specified or required temperatures within plus or minus 2-1/2°F. Each such switch shall be capable of operation at rated voltage when inclined at an angle of 45 degrees from the vertical in any direction and not malfunction when inclined at an angle of 60 degrees from the vertical in any direction. Contactors shall have sufficient capacity to interrupt, at rated voltage, currents equal to ten times the contactor rating. Thermostatic switches shall be fitted with watertight enclosures.

3.5.4 Wiring. - Wiring between motor, switches, cut-offs, and other controlling or heating appliances attached to the machine shall be installed in electrical metallic tubing and suitably connected. Wiring on the machine to be connected to devices beyond the machine, such as the control console, shall terminate in an outlet box or boxes for conduit connection to the devices beyond the machine.

3.5.5 Heaters. - Heaters shall be of the immersion type of sufficient capacity to maintain specified temperatures. Wire in the elements shall be nickel-chromium alloy. Casing surrounding the elements shall be corrosion - resistant and water tight. At rated voltage, the watts input shall be within 5 percent of the heater's nominal rated capacity. Heaters having ratings of 1000 watts and above shall operate on 440 volt, 3 phase, 60 cycle, alternating current, heaters with ratings below 1000 watts may operate on 115 volt, 3 phase, 60 cycle (but not single phase), alternating current, unless otherwise specified (see 6.1).

3.6 Racks, dishwashing. - Four plate racks and four combination racks for cups, bowls and mixed dishes shall be furnished with each machine. The racks shall be suitable for use with the machine supplied. The racks shall either consist of stainless steel wire frames with plastisol coated wire inserts or be molded construction of polypropylene.

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3.7 Technical manuals. - Technical manuals shall be of a commercial type, consisting in general of the manufacturer's standard literature, illustrations or drawings to concisely outline the proper operation, maintenance and parts identification of the machine furnished. Each manual shall be suitably bound in a self-cover of durable quality. Two copies shall be submitted to the ordering activity for assignment of a NAVSHIPS number. Two copies of the approved manual shall be packed with each machine. Additional copies of the approved manual for stock shall be as specified (see 6.1).

3.8 Repair parts. - Repair parts shall accurately fit the machine for which they are specified, and shall be interchangeable with machines of the same type. Repair parts shall be furnished, in general, as follows, or as determined upon completion of Form 4786 (Copies of form 4786 may be obtained from the cognizant Naval Inspector).

3.8.1 Mechanical. - One set of mechanical onboard repair parts shall be furnished with each machine as follows:

1 set of thermometers.
1 thermostatic switch of each type or size used.
1 pump impeller with necessary bearings and seals.
1 set of motor bearings.

3.8.2 Electrical. - Electrical onboard repair parts shall consist of one set each, for each size component used, of those items subject to normal wear or breakdown, such as starter coils; solenoid coils; and controller parts subject to wear, such as line contactors.

3.9 Identification plate. - An identification plate of durable material shall be secured in a conspicuous place on each machine. Marking shall be of a method which will be permanent and legible. The plate shall include information relative to the manufacturer's name, type, serial number, tank capacity in gallons, date of manufacture and contract number, electrical characteristics and the applicable Federal stock number (when given in contract or order).

3.10 Instruction plate. - Each machine shall have an instruction plate made form durable material that is attached to the machine in a location readily visible to the operator. Marking shall be permanent and legible. The instruction plate shall briefly outline operating instructions, including proper water temperatures.

3.11 Workmanship. - The design, materials of construction, fabrication, finish and operation of the machine shall be suitable for the purpose intended and free of all flaws that may affect appearance, sanitation and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory

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acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction. - Prior to the start of production and within 45 days after the date of award of the contract or order, one machine, complete with all parts and accessories, under the contract or order shall be submitted to the U.S. Naval Supply Research and Development Facility, Bayonne, New Jersey, for tests 4.2.3.1 through 4.2.3.5. The preproduction sample is defined as a production line model of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the preproduction sample by the U.S. Naval Supply Research and Development Facility will not relieve the contractor of the responsibility of compliance with the requirements of this specification. Any change or deviations from the preproduction sample shall be submitted for the approval of the bureau or agency concerned. The approved preproduction sample will be used as a guide in the inspection of production units, and shall be presented for delivery as the last unit for delivery under the contract.

4.2.1 Prior to submitting the preproduction sample for testing, the manufacturer shall subject the machine to the test specified in 4.2.3.3. Certification containing actual test data shall be furnished showing that the sample submitted for preproduction tests has successfully passed the test specified in 4.2.3.3.

4.2.2 When a supplier has previously supplied equipment on a recent contract or order, identical to that covered by this specification, which has passed the preproduction tests specified herein, the requirements for preproduction tests may be waived by the bureau or agency concerned, provided acceptable test data is furnished.

4.2.3 Preproduction tests.-

4.2.3.1 Vibration and impact tests. - The dishwashing machine shall be tested without packaging or crating. The vibration test shall be performed prior to the impact test. The dishwashing machine tested shall be capable of performing its required functions after testing.

4.2.3.1.1 Vibration test. - The dishwashing machine shall be placed on the table of a vibration tester. Fences, suitable for the dishwashing machine being tested, may be attached to the test table. While the machine is operating (means to be provided by laboratory to keep machine cycling) without dishes, it shall be vibrated so that it leaves the table momentarily at some interval during the vibration cycle. The proper vibration frequency shall be such that a 1/16 inch thick shim may be inserted between one bottom edge of the machine and the table of the vibration machine. The dishwashing machine shall be vibrated for 2 hours. The function of this test is to produce a measured oscillation upon the equipment to determine its ability to withstand dynamic forces due to vibration which may be encountered aboard ship. The dishwashing machine shall function satisfactorily during this test.

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4.2.3.1.2 Impact test. - An inclined impact testing device similar to the Conbur machine, or equal equipment producing equivalent results, and a shock recorder shall be used. The dishwashing machine shall be fastened to the base so that the dishwashing machine will make no physical contact with the barrier of the impact machine. The assembly shall be placed on the dolly with the face or edge of the base which is to receive the impact projecting approximately 2 inches beyond the forward end of the dolly. The shock recorder shall be positioned on the machine to record the maximum shock received during the impact test. The dolly shall be drawn up the incline to a position which will produce an impact which has a vertical component equivalent to 10 gravity.

4.2.3.2 Overall construction and performance. - The dishwashing machine shall be tested and inspected to determine compliance with the requirements of 3.3.1 through 3.6, and 3.11.

4.2.3.3 Insulation. - The insulation resistance of power carrying electrical components shall be measured by a 500 volt, d.c., insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.2.3.4 Functional test. - This test shall also include the dishwashing machine racks, furnished with the machine, to determine that the design of the racks is suitable for the purpose intended, that is, masking of the items does not occur. Determine that the machine will satisfactorily and uniformly wash and sanitize one rack each of dinner plates, cups and bowls when the food holding surface of the respective items has been coated with the specified test mixture. The ingredients shall be mixed for 3 minutes at low speed in a Waring type blender, the mixture brushed on the dishes and the dishes allowed to stand approximately 30 minutes prior to being placed in the dishwashing machine. The test mixture shall be composed of the following:

8 ounces of peanut butter
8 ounces of salad oil
1/4 ounce of food coloring
9 ounces of flour
17 ounces of water

4.2.3.5 Phase unbalance. - Power unbalance between phases of heaters shall not exceed 10 percent.

4.3 Material certification. - The contractor shall certify in writing that the materials used in the construction of the dishwashing machines furnished conform to the material requirements specified herein. Such certification shal document actual tests, examination or other verifiable quality data.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Dishwashing machine and detached parts.-

5.1.1.1 Preservation and packaging. - Each dishwashing machine and detached parts shall be preserved and packaged in accordance with the supplier's commercial practice to provide protection from hazards of contamination and physical damage in general handling, storage and issue.

5.1.1.2 Packing. - Each dishwashing machine and detached parts shall be packed in containers in such a manner as to insure acceptance by common carrier and safe delivery at destination. Containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include name of item; type of item, Federal stock number (if provided in contract or order); contract or order number; contractor's name; and the destination.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Voltage for motors if other than specified (see 3.5).
(c) Voltage for heaters if other than specified (see 3.5.5).
(d) Quantity of manuals for stock (see 3.7).
(e) Repair parts, if other than required (see 3.8).
(f) Preservation, packaging, packing or marking requirements if other than specified (see 5.1).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-24194(SHIPS)	DISHWASHING MACHINE, COMMERCIAL (SUBMARINE USE)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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